

Towards Designing the Sense of Touch: Innovative Tools for Haptic Feedback in Extended Reality

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ABSTRACT

Extended Reality (XR) technologies rapidly evolve, offering remarkable opportunities for creating immersive and interactive experiences. However, the integration of effective haptic feedback remains a significant challenge. We aim to explore and enhance haptic experience design based on three major themes: wearable hardware, rendering methods, and design systems incorporating algorithms. Through this comprehensive approach, we seek to improve the quality of haptic interactions in XR environments significantly.

CCS CONCEPTS

• **Human-centered computing** → **Systems and tools for interaction design**; **User interface toolkits**; **Haptic devices**; **Virtual reality**.

KEYWORDS

Haptics Design, Authoring System, XR, HCI

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1 INTRODUCTION

The rapid advancement of XR technologies has opened up new frontiers in creating immersive and interactive experiences. However, a critical challenge remains in effectively integrating haptic feedback to enhance user engagement and realism. Traditional haptic rendering methods are often limited and lack the sophistication needed for truly immersive XR environments. Our proposed research aims to develop advanced haptic experiences using a new haptic rendering method, an authoring tool for designers, and AI-driven haptic generation to bridge the gap between haptic technology and immersive XR applications.

Haptic Experience Design (HaXD) requires a comprehensive understanding of hardware and software, unlike conventional User Experience (UX) techniques. Currently, the lack of haptic designers and high barriers interrupt the development of various realistic

tactile content within XR. To address it, a sufficient discussion of the scalability and freedom of the haptic design process is needed. Furthermore, an agreed-upon design framework of haptic researchers may improve the accessibility of designing haptic experiences.

2 RELATED WORK

2.1 Wearable Haptic Device for Hands

Commercial haptic gloves with compact form factors have been produced to provide vibrotactile input to users' hands [1, 14]. However, because of the insufficient use of tactile illusions and a large reliance on single-point vibrations, these gloves provide tactile feedback to the hand with limited spatial coverage. Recently, researchers proposed glove-based whole-hand vibrotactile feedback [4, 5, 10, 11, 16] to give direct hand stimulation and diverse interactive experiences. These studies demonstrate increased spatiotemporal coverage and perceived performance of tactile sense throughout the entire hand. However, a static hand posture for displaying tactile experience limits feasibility and scalability.

2.2 Rendering Techniques for Immersive Haptics

Changing hand postures easily changed the haptic perception of tactile sensation [8, 15]. For instance, even a tiny change in hand configuration might alter a human's perception of vibrotactile feedback [17]. Hand position affects signal propagation and the direction of a succession of stimuli. Without regard for hand positions, haptic feedback frequently produces erroneous feelings, impairing both perceptual recognition performance and immersion. Therefore, algorithms and rendering methods that operate wearable devices are essential topics in the tactile experience.

2.3 Haptic Design in HCI

Recent works include a responsive web-based authoring interface [13], trace-based vibrotactile feedback design [1], and 3D body sketching [2] to facilitate interactive haptic design. These methods enable instant haptic input and fine-tuning control panels, enabling for faster design revisions. However, a 2D desktop GUI-based toolkit makes it difficult to achieve in-situ haptic design alterations for VR hand interactions since users must transition between the desktop and VR interfaces to create a haptic impact. Recent research suggests in-situ prototyping for a VR environment with direct manipulation [3, 6]. Inspired by previous studies, we extend an in-situ VR haptic design tool that allows users to construct vibrotactile feedback patterns on their hands with various hand postures.



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3 RESEARCH GOALS

To establish the research goal, we highlight the limits of existing haptic experience designs and aim to investigate the conditions and methodologies required for creating simple tactile experiences. Existing haptic experience designs rely on the inventiveness of developers (Haptician), making it time-consuming to achieve haptic outputs that represent user goals.

3.1 Haptic Glove with Sensation

To address current challenges, we have focused on combining advanced technology with user-centered design principles. As part of this effort, we devised a haptic glove to deliver the most immersive tactile feedback possible. We developed a haptic glove with 13 motors to augment users' movement while providing rich sensations (Figure 1). To determine the number of motors, we looked at the required distance between them. The motors should be close enough to provide phantom sensation while maintaining a sufficient distance to prevent disturbance and pain during hand movements [9, 12].

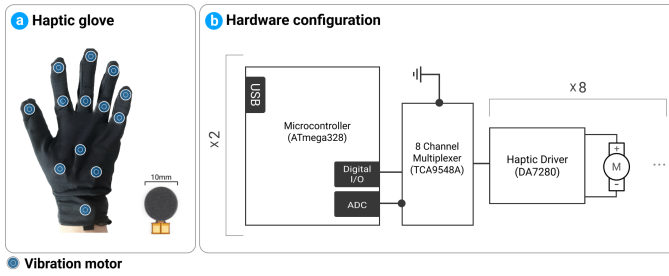


Figure 1: Our proof-of-concept haptic glove embedded with 13 LRAs and hardware configuration.

Sports gloves (85% polyester, 15% polyurethane) were used; they are spandex-based and form-fitting, preserving hand shape for stable hand tracking with VR head-mounted displays (HMD). Inside the glove, 13 LRAs (VG1040003D, Vybronic) are affixed using double-sided adhesive (3M, 468MP). To control a total of 13 haptic drivers (DA7280, Dialogue), we used two Sparkfun Redboards (Atmega328) with 8-to-1 multiplexers (TCA9548A, Texas Instrument) (Figure 1b). To ensure strong connections during hand movements, we allowed interconnecting wires to have a length of around ≥ 30 cm. The maximum frame rate was 250 Hz, with a negligible latency of less than 4 ms between the command and actuation trigger.

In order to provide continuous vibrotactile feedback beyond basic point and line spanning the entire hand, we incorporated phantom tactile sensation techniques [7, 12]. We combined 1D and 2D phantom sensation techniques with a unique actuator arrangement to enable a rich haptic experience covering the whole hand. We developed haptic rendering methods in accordance with the phantom grid of the target posture to maintain similar vibrotactile sensations under different hand postures.

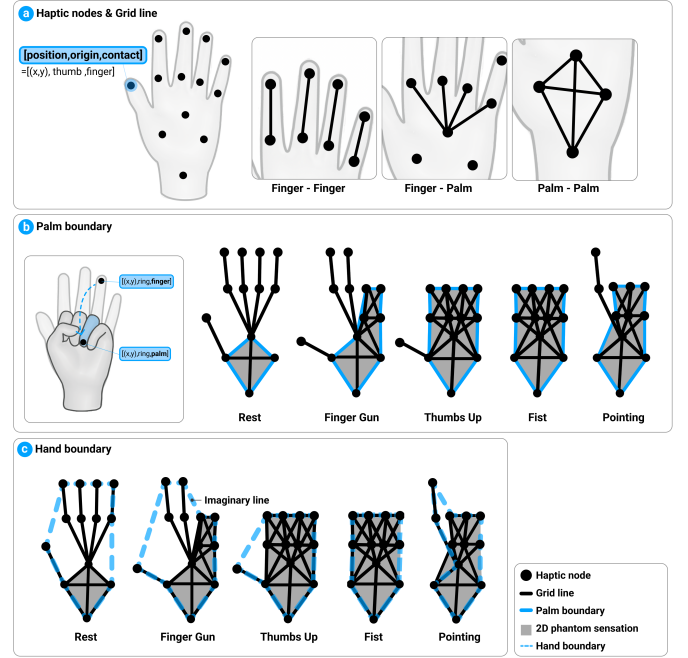


Figure 2: Overview of phantom grid.

3.2 Dynamic Haptic Rendering Method

In addition to hardware, we need to carefully consider the methods used for rendering as well. As a result, we proposed a novel abstraction for haptic design, which is made up of lines and nodes that together capture the spatial information of the entire hand. Creating a haptic rendering map to use 1D and 2D phantom sensations is the primary function of the phantom grid. The main elements of the phantom grid construction are shown in Figure 5.

With the suggested phantom grid and phantom sensation methods, our tool enables haptic rendering for various hand postures. Haptic traces are 2D traces that the users build to produce vibrotactile patterns. We referred to the spots in these traces—which were collected at 100 Hz—as haptic points. We implemented different haptic rendering techniques based on the geometric correspondence between the phantom grid and the haptic point. The geometric connection consists of the following: inside-palm localization, outside-palm localization, grid line intersection, and haptic node collision. In the haptic experience evaluation, the proposed posture-adaptive rendering algorithm achieved comparable information transmission (IT), recognition accuracy, and similarity scores in our user studies with 20 participants.

3.3 Designing User Experience with Toolkit

To ensure the finest possible haptic creation, we designed the entire process with a better user experience in mind. We created an in-situ VR authoring framework that offers 1) high levels of customization, 2) great design freedom with a reliable system, and 3) automated posture-adaptive haptic rendering, taking into account the conclusions from the aforementioned investigations.

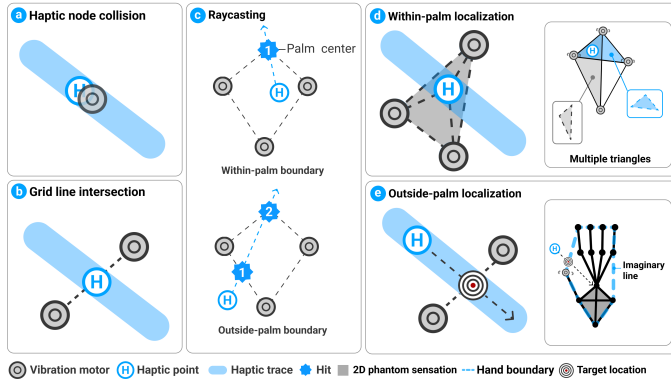


Figure 3: Posture-adaptive haptic rendering methods.

Users can sketch with the virtual hand set in the chosen hand posture. *Rest*, *Finger Gun*, *Thumbs Up*, *Fist*, and *Pointing* are the five hand postures that we supported. The designer can draw multiple separate traces that can overlap in space. We extended the canvas's boundaries to include space beyond the hand (Figure 4d) to allow users additional design flexibility. A part of the system that allows for instant playback involves animating haptic traces on a virtual hand that is empty (Figure 4a~c). It is simple for hapticians to experience designed vibrotactile feedback in-situ since the animated haptic traces are synchronized with real hardware. Finally, by using the "save" option, the haptic design can be preserved. This will cause our posture-adaptive algorithm to apply to the produced design, automatically producing vibrotactile patterns for various hand positions. The UI panel allows the designers to choose various hand positions, which automatically modifies the haptic design. Additionally, hapticians can modify the haptic design for any posture as needed.

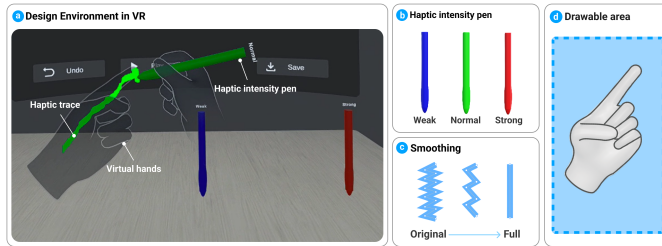


Figure 4: User interfaces of the haptic authoring toolkit in VR.

4 POTENTIAL RESEARCH DIRECTION

Based on previous research, there is a great chance to expand the scope of the authoring environment by developing a user-friendly design process and presenting an authoring system that enables immediate haptic experience design in VR.

AI Haptic Design Tool for Designers Voice, graphics, video, and 3D content can be generated according to user input data (e.g., text), while discussions on the tactile experience are still insufficient.

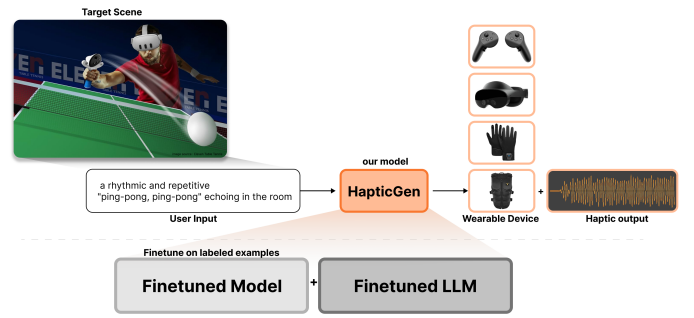


Figure 5: Overview of the AI Haptic Design approach.

A haptic dataset is required to train a generative AI model, but there is no published haptic dataset. This is due to the fact that the agreed design framework and authoring system that can be designed have not been developed in common. To solve the problem, we intend to explore the possibilities of converting training data used by other modalities into haptic-specialized data.

The AI tool works as follows:

- **Data collection:** Collecting haptic data aligns with existing XR and user interactions to create a comprehensive dataset.
- **Extracting Features:** Using this dataset, the tool identifies common patterns and optimal haptic feedback combinations.
- **Design suggestions:** The AI system suggests haptic feedback designs that match the desired user interaction, which designers can customize further.

5 EXPECTED CONTRIBUTION

Through academic performance related to Human-Computer Interaction (HCI), research designs and approaches can be devised for developing tactile experiences in virtual environments that resemble reality. The ultimate goal is to develop innovative technologies and contribute to the XR industry through interdisciplinary research between science and technology, humanities, and sociology. In an effort to push the boundaries of haptic design, machine learning algorithms are being used to propose the best haptic feedback patterns depending on the context and content of the XR environment. Designers can input parameters such as the type of interaction (e.g., grabbing, touching, or swiping) and the AI tool will generate haptic feedback patterns that enhance the user's immersive experience.

This research direction aims to advance the field of XR and haptic technology by developing new tools to simplify the creation and integration of advanced haptic feedback. The haptic authoring tool offers a user-friendly platform for designing detailed haptic experiences, while the AI-driven design tool leverages machine learning to enhance the design process. These tools have the potential to significantly improve the quality of immersive experiences in XR applications, making them more engaging and realistic for users. By addressing the challenges of haptic feedback integration, this research paves the way for more sophisticated and immersive XR environments, ultimately contributing to the broader adoption and development of XR technologies in various fields such as gaming, education, and virtual training.

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A BIOGRAPHICAL SKETCH

Youjin Sung is an early-stage PhD student at KAIST, specializing in Human-Computer Interaction (HCI). She began her doctoral program in 2023 under Professor Sang Ho Yoon and has shown significant dedication to advancing **Haptics**. Her goal is to devise cutting-edge technologies, including an AI-driven haptic design tool and a haptic authoring toolkit. Through her research, she actively addresses possible problems with users. As a result, she has had the opportunity to be awarded and give presentations at numerous HCI conferences.

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